

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8/2	(a)			0.42 (2) if answer incorrect award (1) for any of following 42 420000 4.2×10^5		2		2	2	
	(b)			award (1) for any of following • distillation • reverse osmosis • graphene membrane neutral answers desalination boiling / evaporation / condensation	1			1		
	(c)			add soap solution to both water samples and shake (1) fair testing - equal volumes of both water samples and shake for an equal amount of time (1) the sample that produces more lather is the softer water / the sample that produces less lather is the harder (1) alternative method add 1cm ³ / small volume of soap solution and shake; if no lather add further 1cm ³ / small volume (1) fair testing - equal volumes of both water samples and shake for an equal amount of time (1) the sample that lathers with less soap is the softer / the sample that requires more soap is the harder (1)	3			3		3

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		2Ca + O₂ → 2CaO ☒ Ca + O₂ → 2CaO ☐ Ca + O → CaO ☐		1		1		
				Question 1 total	0	8	0	8	0	0